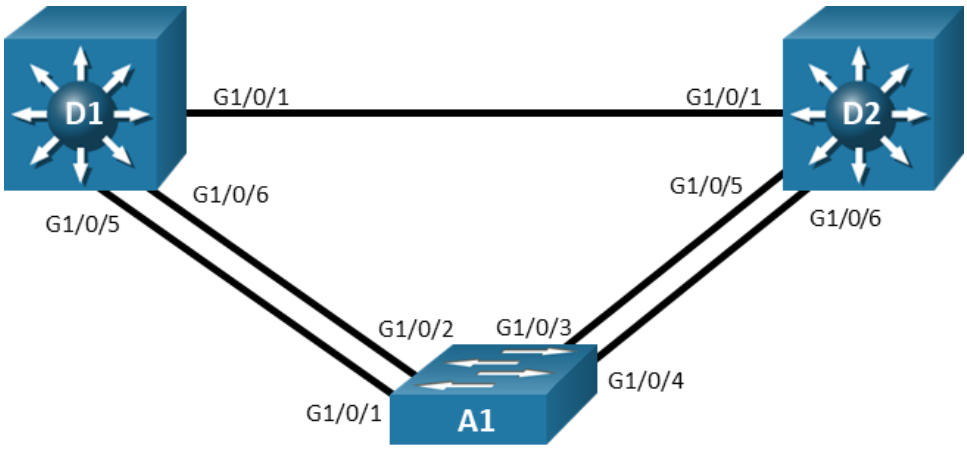
Lab - Implement MST

Topology



Objectives

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

Part 2: Implement and Observe MST

Part 3: Configure, Tune and Verify Basic MST Operation

Background / Scenario

Cisco's Per VLAN Spanning Tree (PVST+) allows each VLAN to run its own independent spanning tree, improving link utilization and network flexibility. However, PVST+ creates a separate spanning tree instance for every VLAN, which can strain switch CPU and memory resources, especially if the number of VLANs exceeds the supported PVST+ instances. This would leave some VLANs unprotected from loops. Additionally, PVST+ and Rapid PVST+ are Cisco-proprietary protocols and are generally not compatible with switches from other vendors.

Multiple Spanning Tree (MST), standardized as IEEE 802.1s and later merged into 802.1Q, is an open protocol derived from RSTP and supports rapid convergence. MST allows multiple VLANs to be mapped to a single spanning tree instance, balancing flexibility and resource efficiency. It is the only standardized spanning-tree protocol for VLAN-based networks, making it suitable for mixed vendor environments.

In this lab, you will set up two instances of MST, one for VLANs 2 and 3 and the other for VLANs 4 and 5. All other VLANs will be mapped to the default MST instance (also referred to as IST or Internal Spanning Tree).

Note: This lab is an exercise in deploying and verifying MST and does not reflect networking best practices.

Note: The switches used with CCNP hands-on labs are Cisco 9300L with Cisco IOS XE release 17.12.05 (universalk9 image) and Cisco 9200L with IOS XE release 17.12.05 (universalk9 image). Other routers and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: Ensure that the switches have been erased and have no startup configurations. If you are unsure contact your instructor.

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Required Resources

- 2 Switches (Cisco 9300L with Cisco IOS XE release 17.12.05 universal image or comparable)
- 1 Switch (Cisco 9200L with Cisco IOS release 17.12.05 universal image or comparable)
- 1 PC (Windows with a terminal emulation program, such as Tera Term)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Instructions

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

In Part 1, you will set up the network topology and configure basic settings.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram and cable as necessary.

Step 2: Configure basic settings for each switch.

a. Console into each switch, enter global configuration mode, and apply the basic settings. The startup configurations for each device are provided below.

Switch D1

```
hostname D1
banner motd # D1, Multiple Spanning Tree #
spanning-tree mode rapid-pvst
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
 exit
interface range g1/0/1, g1/0/5-6
 switchport mode trunk
 no shutdown
 exit
vlan 2
 name SecondVLAN
 exit
vlan 3
 name ThirdVLAN
 exit
vlan 4
 name FourthVLAN
 exit
```

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```
vlan 5
 name FifthVLAN
 exit
```

Switch D2

```
hostname D2
banner motd # D2, Multiple Spanning Tree #
spanning-tree mode rapid-pvst
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
 exit
interface range g1/0/1, g1/0/5-6
 switchport mode trunk
 no shutdown
 exit
vlan 2
 name SecondVLAN
 exit
vlan 3
 name ThirdVLAN
 exit
vlan 4
 name FourthVLAN
 exit
vlan 5
 name FifthVLAN
 exit
```

Switch A1

```
hostname A1
banner motd # A1, Multiple Spanning Tree #
spanning-tree mode rapid-pvst
!Configure A1 to not be the root for all VLANs.
spanning-tree vlan 1-5 priority 61440
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
 exit
interface range g1/0/1-4
```

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```
 switchport mode trunk
 no shutdown
vlan 2
 name SecondVLAN
 exit
vlan 3
 name ThirdVLAN
 exit
vlan 4
 name FourthVLAN
 exit
vlan 5
 name FifthVLAN
 exit
```

b. Set the clock on each switch to UTC time.

c. Save the running configuration to startup-config.

Note: Outputs and Spanning Tree topologies highlighted in this lab may be different than what you observe using your own equipment. It is critically important for you to understand how Spanning Tree makes its decisions, and how those decisions impact the operational topology of the network.

Part 2: Implement and Observe MST

In this Part, we will configure MST on D1 and D2 only. We will do this so we can observe their interaction with each other as well as their interaction with A1.

a. On D1 and D2, issue the command **spanning-tree mode mst**.

```
D1(config)# spanning-tree mode mst
Warning: Changing STP mode can disrupt the traffic and make system unstable
Recommend to change STP mode only during maintenance window
```

b. At this point, with no MST-specific configuration, MST Instance 0 is operational for all VLANs. Issue the command **show spanning-tree** and you will see in the output that the spanning tree information is about MST 0. Issue the command **show spanning-tree mst** and you will see the MST-specific STP information that is specific to MST 0 only. Take note of the information displayed for interfaces g1/0/5 and g1/0/6 because they are connected to a switch that is not running MST. Their type is listed as P2p Bound (PVST).

```
D1# show spanning-tree

MST0
  Spanning tree enabled protocol mstp
  Root ID    Priority    32768
             Address     f8e9.4f9b.7f80
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32768  (priority 32768 sys-id-ext 0)
             Address     f8e9.4f9b.7f80
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
```

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```
Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi1/0/1             Desg FWD 20000     128.1    P2p
Gi1/0/5             Desg FWD 20000     128.5    P2p Bound(PVST)
Gi1/0/6             Desg FWD 20000     128.6    P2p Bound(PVST)

D1# show spanning-tree mst

##### MST0    vlans mapped:   1-4094
Bridge        address f8e9.4f9b.7f80  priority      32768 (32768 sysid 0)
Root          this switch for the CIST
Operational   hello time 2 , forward delay 15, max age 20, txholdcount 6
Configured    hello time 2 , forward delay 15, max age 20, max hops    20

Interface                       Role Sts Cost      Prio.Nbr Type
----------------                ---- --- --------- -------- ------------------------
Gi1/0/1                         Desg FWD 20000     128.1    P2p
Gi1/0/5                         Desg FWD 20000     128.5    P2p Bound(PVST)
Gi1/0/6                         Desg FWD 20000     128.6    P2p Bound(PVST)
```

c. The basic behavior of MST is the same as spanning tree, where a root bridge must be selected, then root ports, and finally best paths to the root bridge from all non-root bridges. In the current network, we can see that D1 has been elected the root bridge. The bridge priority defaults to 32768, so the election is based on D1 having a lower base MAC address. The switch elected as root may be different in your lab topology, but the rules for election remain the same. Issue the command **show spanning-tree root** on switch A1. Switch A1 is running five instances of spanning tree.

```
A1# show spanning-tree root

                                        Root    Hello Max Fwd
Vlan                Root ID             Cost    Time  Age Dly  Root Port
---------------- ------------------- --------- ----- --- ---  ------------
VLAN0001         32768 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1
VLAN0002         32768 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1
VLAN0003         32768 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1
VLAN0004         32768 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1
VLAN0005         32768 f8e9.4f9b.7f80     20000    2   20  15  Gi1/0/1
```

d. Issue the **show spanning-tree root** command on D2 and the output will be different. This is because with MST, only one instance of the spanning-tree algorithm runs, regardless of the number of VLANs mapped to it.

```
D2# show spanning-tree root

                                        Root    Hello Max Fwd
MST Instance        Root ID             Cost    Time  Age Dly  Root Port
---------------- ------------------- --------- ----- --- ---  ------------
MST0             32768 f8e9.4f9b.7f80         0    2   20  15  Gi1/0/1
```

e. Configure A1 to use MST.

```
A1(config)# spanning-tree mode mst
```

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Part 3: Configure, Tune and Verify Basic MST Operation

In the last part, you configured all three switches to run MST. In this part, you will further configure, tune, and verify MST to support the unique topological requirements.

Step 1: Create and verify an MST configuration.

MST allows network engineers to reduce the load of the spanning-tree protocol while still providing unique spanning-tree topologies for groups of VLANs. MST configuration must be completed by hand on each switch individually.

Complete the following tasks on switch D1:

a. Enter MST configuration mode using the command **spanning-tree mst configuration.**

```
D1(config)# spanning-tree mst configuration
```

b. Configure an MST region name. Our example will be CCNP.

```
D1(config-mst)# name CCNP
```

c. Configure an MST configuration revision number. Our example will be 1.

```
D1(config-mst)# revision 1
```

d. Configure instance 1 to include VLAN 2.

```
D1(config-mst)# instance 1 vlan 2
```

e. Configure instance 2 to include VLAN 4.

```
D1(config-mst)# instance 2 vlan 4
```

f. Commit the configuration by typing exit and returning to global configuration mode.

```
D1(config-mst)# exit
D1(config)# end
```

g. Issue the command **show spanning-tree mst** to verify the configuration is in place. MST1 should show VLAN 2 mapped. MST2 should show VLAN 4 mapped. MST0 should show all the remaining VLANs mapped. Under the Type column, **P2p Bound(RSTP)** means that link is point-to-point, and D1 has detected the neighbor is not speaking MST (802.1s) but IEEE RSTP (802.1w). So, the port is operating as an MST region boundary and uses RSTP compatibility on that link.

```
D1# show spanning-tree mst

##### MST0    vlans mapped:   1,3,5-4094
Bridge        address f8e9.4f9b.7f80  priority      32768 (32768 sysid 0)
Root          address 4464.3ce7.6880  priority      32768 (32768 sysid 0)
              port    Gi1/0/5         path cost     20000
Regional Root this switch
Operational   hello time 2 , forward delay 15, max age 20, txholdcount 6
Configured    hello time 2 , forward delay 15, max age 20, max hops    20

Interface                     Role Sts Cost      Prio.Nbr Type
----------------              ---- --- --------- -------- ----------------
Gi1/0/1                       Altn BLK 20000     128.1    P2p Bound(RSTP)
Gi1/0/5                       Root FWD 20000     128.5    P2p Bound(RSTP)
Gi1/0/6                       Altn BLK 20000     128.6    P2p Bound(RSTP)

##### MST1    vlans mapped:   2
```

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```
Bridge        address f8e9.4f9b.7f80  priority      32769 (32768 sysid 1)
Root          this switch for MST1

Interface                       Role Sts Cost      Prio.Nbr Type
----------------                ---- --- --------- -------- ----------------
Gi1/0/1                         Altn BLK 20000     128.1    P2p Bound(RSTP)
Gi1/0/5                         Mstr FWD 20000     128.5    P2p Bound(RSTP)
Gi1/0/6                         Altn BLK 20000     128.6    P2p Bound(RSTP)

##### MST2     vlans mapped:   4
Bridge        address f8e9.4f9b.7f80  priority      32770 (32768 sysid 2)
Root          this switch for MST2

Interface                       Role Sts Cost      Prio.Nbr Type
----------------                ---- --- --------- -------- ----------------
Gi1/0/1                         Altn BLK 20000     128.1    P2p Bound(RSTP)
Gi1/0/5                         Mstr FWD 20000     128.5    P2p Bound(RSTP)
Gi1/0/6                         Altn BLK 20000     128.6    P2p Bound(RSTP)
```

h. This configuration does not propagate to other switches. Each switch exchanges digest information summarizing the VLAN-to-Instance mappings it has configured. If a switch receives a BPDU with a different digest, it assumes that the sender is in a different MST region. On A1, confirm that it is not in an MST region with D1. Notice A1 only has the default MST0 instance and that the ports connected to D1 are classified as **P2p**.

Note: Seeing **Bound(RSTP)** on D1 and only **P2p** on A1 is normal and implementation-dependent. Your lab setup may show G1/0/1 and G1/0/2 as **Bound(RSTP)**.

```
A1# show spanning-tree mst

##### MST0      vlans mapped:   1-4094
Bridge        address 4464.3ce7.6880  priority      32768 (32768 sysid 0)
Root          this switch for the CIST
Operational   hello time 2 , forward delay 15, max age 20, txholdcount 6
Configured    hello time 2 , forward delay 15, max age 20, max hops    20

Interface                       Role Sts Cost      Prio.Nbr Type
----------------                ---- --- --------- -------- ----------------------
Gi1/0/1                         Desg FWD 20000     128.1    P2p
Gi1/0/2                         Desg FWD 20000     128.2    P2p
Gi1/0/3                         Desg FWD 20000     128.3    P2p
Gi1/0/4                         Desg FWD 20000     128.4    P2p
```

You can also confirm that D1 and A1 have different MST configuration digests by using the **show spanning-tree mst configuration digest** command.

```
A1# show spanning-tree mst configuration digest
Name      []
Revision  0     Instances configured 1
Digest          0xAC36177F50283CD4B83821D8AB26DE62
Pre-std Digest  0xBB3B6C15EF8D089BB55ED10D24DF44DE
```

And here is the digest from D1:

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```
D1# show spanning-tree mst configuration digest
Name      [CCNP]
Revision  1     Instances configured 3
Digest          0x746D865FEAD726D8F401F9396B8B62DA
Pre-std Digest  0xDE5D7C8B79A99142EBC0A1C265ED7B05
```

As you can see here, the digest values are different.

i. Configure MST on D1, change the revision number to 2 and add VLAN 3 to instance 1 and VLAN 5 to instance 2. The configuration does not take affect until you exit MST configuration mode.

```
D1(config)# spanning-tree mst configuration
D1(config-mst)#  show current
Name      [CCNP]
Revision  1     Instances configured 3

Instance  Vlans mapped
--------  ---------------------------------------------------------------------
0         1,3,5-4094
1         2
2         4
---------------------------------------------------------------------------
D1(config-mst)# revision 2
D1(config-mst)# instance 1 vlan 3
D1(config-mst)# instance 2 vlan 5
D1(config-mst)# show pending
Pending MST configuration
Name      [CCNP]
Revision  2     Instances configured 3

Instance  Vlans mapped
--------  ---------------------------------------------------------------------
0         1,6-4094
1         2-3
2         4-5
---------------------------------------------------------------------------
D1(config-mst)# exit
D1(config)# end
```

j. Now configure D2 and A1 with the same configuration settings (name CCNP, revision2, instance 1 vlans 2-3, instance 2 vlans 4-5). After completing the configuration on D2 and A1, the output of **show spanning-tree mst** on A1 should be similar to the following output. Notice that, even though we configured A1 at the beginning of this lab to never be the root with the **spanning-tree vlan 1-5 priority 61440** command, this configuration does not apply to MST. In our test, A1 has the better Bridge ID so it becomes root for all MST instances. Your results may differ.

```
A1# show spanning-tree mst

##### MST0    vlans mapped:   1,6-4094
Bridge        address 4464.3ce7.6880  priority      32768 (32768 sysid 0)
Root          this switch for the CIST
Operational   hello time 2 , forward delay 15, max age 20, txholdcount 6
```

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```
Configured    hello time 2 , forward delay 15, max age 20, max hops    20

Interface                       Role Sts Cost      Prio.Nbr Type
----------------                ---- --- --------- -------- --------------------
Gi1/0/1                         Desg FWD 20000     128.1    P2p
Gi1/0/2                         Desg FWD 20000     128.2    P2p
Gi1/0/3                         Desg FWD 20000     128.3    P2p
Gi1/0/4                         Desg FWD 20000     128.4    P2p

##### MST1    vlans mapped:   2-3
Bridge        address 4464.3ce7.6880  priority      32769 (32768 sysid 1)
Root          this switch for MST1

Interface                       Role Sts Cost      Prio.Nbr Type
----------------                ---- --- --------- -------- --------------------
Gi1/0/1                         Desg FWD 20000     128.1    P2p
Gi1/0/2                         Desg FWD 20000     128.2    P2p
Gi1/0/3                         Desg FWD 20000     128.3    P2p
Gi1/0/4                         Desg FWD 20000     128.4    P2p

##### MST2    vlans mapped:   4-5
Bridge        address 4464.3ce7.6880  priority      32770 (32768 sysid 2)
Root          this switch for MST2

Interface                       Role Sts Cost      Prio.Nbr Type
----------------                ---- --- --------- -------- --------------------
Gi1/0/1                         Desg FWD 20000     128.1    P2p
Gi1/0/2                         Desg FWD 20000     128.2    P2p
Gi1/0/3                         Desg FWD 20000     128.3    P2p
Gi1/0/4                         Desg FWD 20000     128.4    P2p
```

Step 2: Controlling the Root Bridge.

Just as in PVST+ and Rapid PVST+, the root bridge for each MST instance is elected based on the lowest Bridge ID. When priorities and extended system IDs are identical, the switch with the numerically lower MAC address becomes root. In an unconfigured network, one switch will be elected as the root bridge, which may or may not be desirable.

With MST, root bridge selection is based on instances.

Controlling Root Bridge Location

There are two ways to set the root bridge:

- Use **spanning-tree mst** *instance-id* **priority** *value* to set a specific priority value.
- Use **spanning-tree mst** *instance-id* **root** { **primary** | **secondary** } to have the switch set an appropriate priority automatically.

Behavior of Each Command

- The **priority** command sets an explicit priority value (a multiple of 4096).
- The **root primary** command sets the local bridge's priority to 24,576 if its MAC is lower than the current root's, or to 4,096 less than the current root's priority if its MAC is higher.

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The logic is straightforward: root primary lowers the priority only as much as needed to win the election, while leaving the range 24,576 to the default 32,768 available for secondary roots. The calculation always considers the entire Bridge ID.

The **spanning-tree mst** *instance-id* **root secondary** command sets the local bridge's priority to 28,672. In a default network (all switches at 32,768), root primary sets the priority to 24,576 (two increments below default), and root secondary sets it to 28,672 (one increment below default).

a. Modify D1 and D2 so that D1 is elected the primary root bridge for MST Instances 0 and 1 and D2 is elected the primary root bridge for MST Instance 2. D1 should be elected as the secondary root bridge for MST Instance 2, and D2 should be elected as the secondary root bridge for MST Instances 0 and 1. You will need to make configuration changes on both D1 and D2. The commands used at D1:

```
D1(config)# spanning-tree mst 0-1 root primary
D1(config)# spanning-tree mst 2 root secondary
```

b. After you have configured both D1 and D2, go to A1 and issue the command **show spanning-tree root**. In this output, you will see that A1 is no longer the root and that the root bridges are different for each configured MST instance.

```
A1# show spanning-tree root

                                        Root    Hello Max Fwd
MST Instance            Root ID         Cost    Time  Age Dly  Root Port
---------------- -------------------- --------- ----- --- ---  ------------
MST0             24576 f8e9.4f9b.7f80         0     2   20  15  Gi1/0/1
MST1             24577 f8e9.4f9b.7f80     20000     2   20  15  Gi1/0/1
MST2             24578 f8e9.4f9b.8700     20000     2   20  15  Gi1/0/3
```

From the above output, you can see that the root port for MST1 is G1/0/1 and the root port for MST2 is G1/0/3. And we made sure A1 is not the root bridge for MST Instance 0.

Step 3: Adjust port cost values to impact root and designated port selection.

As the network is implemented right now, there are two direct paths between switch A1 and the root bridge for each MST. Path and port costs are evaluated to determine the shortest path to the root bridge. In the case where there are multiple equal cost paths to the root bridge, additional attributes must be evaluated. In our case, the lower interface number (for example, G1/0/1) is chosen as the Root Port, and the higher interface number (for example, G1/0/2) is put into a spanning tree Blocking state.

You can see which ports are blocked with the **show spanning-tree** *vlan-id* command or the **show spanning-tree blockedports** command. For now, examine VLAN 1 on D1.

a. On A1, issue the commands **show spanning-tree vlan 1** and **show spanning-tree blockedports**.

```
A1# show spanning-tree mst 1

##### MST1    vlans mapped:   2-3
Bridge        address 4464.3ce7.6880  priority      32769 (32768 sysid 1)
Root          address f8e9.4f9b.7f80  priority      24577 (24576 sysid 1)
              port    Gi1/0/1         cost          20000     rem hops 19

Interface                     Role Sts Cost      Prio.Nbr Type
----------------              ---- --- --------- -------- --------------------
Gi1/0/1                       Root FWD 20000     128.1    P2p
Gi1/0/2                       Altn BLK 20000     128.2    P2p
Gi1/0/3                       Altn BLK 20000     128.3    P2p
Gi1/0/4                       Altn BLK 20000     128.4    P2p
```

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```
A1# show spanning-tree blockedports

Name                 Blocked Interfaces List
-------------------- ------------------------------------
MST0                 Gi1/0/2, Gi1/0/3, Gi1/0/4
MST1                 Gi1/0/2, Gi1/0/3, Gi1/0/4
MST2                 Gi1/0/1, Gi1/0/2, Gi1/0/4

Number of blocked ports (segments) in the system : 9
```

As you can see, MST 1 has its Root Port on G1/0/1. Ports G1/0/2, G1/0/3, and G1/0/4 are Alternate Blocking Ports.

To manipulate which port becomes the Root Port on non-root bridges, change the port cost (a value between 1 and 200,000,000) or port priority value (a value between 0 and 240 in increments of 16). Remember that this change could have an impact on downstream switches as well.

Note: The changes you are about to implement are considered topology changes and could have a significant impact on the overall structure of the spanning tree in your switch network. Do not make these changes in a production network without careful planning and prior coordination.

b. On A1, shut down interfaces G1/0/1 and G1/0/2, assign a new port cost of 1000 to G1/0/2 using the **spanning-tree mst 1 cost** *value* command, and then issue the **no shutdown** command on the ports.

```
A1# conf t
Enter configuration commands, one per line.  End with CNTL/Z.
A1(config)# interface range g1/0/1-2
A1(config-if-range)# shutdown
A1(config-if-range)# exit
A1(config)# interface g1/0/2
A1(config-if)# spanning-tree mst 1 cost 1000
A1(config-if)# exit
A1(config)# interface range g1/0/1-2
A1(config-if-range)# no shutdown
A1(config-if-range)# end
```

c. Now verify that this impacts root port selection on A1 using the **show spanning-tree vlan 1** and **show spanning-tree blockedports** commands.

```
A1# show spanning-tree mst 1

##### MST1    vlans mapped:   2-3
Bridge        address 4464.3ce7.6880  priority      32769 (32768 sysid 1)
Root          address f8e9.4f9b.7f80  priority      24577 (24576 sysid 1)
              port    Gi1/0/2         cost          1000      rem hops 19

Interface                     Role Sts Cost      Prio.Nbr Type
----------------              ---- --- --------- -------- --------------------------
Gi1/0/1                       Altn BLK 20000     128.1    P2p
Gi1/0/2                       Root FWD 1000      128.2    P2p
Gi1/0/3                       Desg FWD 20000     128.3    P2p
Gi1/0/4                       Desg FWD 20000     128.4    P2p
```

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```
A1# show spanning-tree blockedports

Name                 Blocked Interfaces List
-------------------- ------------------------------------
MST0                 Gi1/0/2, Gi1/0/3, Gi1/0/4
MST1                 Gi1/0/1
MST2                 Gi1/0/1, Gi1/0/2, Gi1/0/4

Number of blocked ports (segments) in the system : 7
```

From the output, you can see that the root port selected by A1 for MST1 is now interface G1/0/2, and the port (and root) cost is now 1000.

Step 4: Adjust port priority values to impact root port selection.

The next method to impact root port selection is configured on the root bridge itself. In our current network topology, A1 has two connections to D2 for MST Instance 2. The root port has been selected, in this case based on the lowest port ID. Port ID is made up of two values, labeled as Prio (Priority) and Nbr (Number).

Note: The port number is not necessarily equal to the interface ID. A switch may use any port number for STP purposes as long as they are unique for each port on the switch.

The port priority can be any value between 0 and 240, in increments of 16 (older switches may allow setting the priority in different increments).

a. On A1, issue the command **show spanning-tree mst 2** and take note of the port ID values listed.

```
A1# show spanning-tree mst 2

##### MST2    vlans mapped:   4-5
Bridge        address 4464.3ce7.6880  priority      32770 (32768 sysid 2)
Root          address f8e9.4f9b.8700  priority      24578 (24576 sysid 2)
              port    Gi1/0/3         cost          20000     rem hops 19

Interface                       Role Sts Cost      Prio.Nbr Type
----------------                ---- --- --------- -------- --------------------------
Gi1/0/1                         Altn BLK 20000     128.1    P2p
Gi1/0/2                         Altn BLK 20000     128.2    P2p
Gi1/0/3                         Root FWD 20000     128.3    P2p
Gi1/0/4                         Altn BLK 20000     128.4    P2p
```

As expected with two equal-cost paths to the root bridge, the lower port ID was selected as the root port.

b. Modify the port priority of D2 interface G1/0/6 so that it becomes the preferred port by issuing the **spanning-tree mst 2 port-priority** *value* interface configuration command. Use a value of 64.

```
D2# config t
Enter configuration commands, one per line.  End with CNTL/Z.
D2(config)# interface range g1/0/5-6
D2(config-if-range)# shutdown
D2(config-if-range)# exit
D2(config)# interface g1/0/6
D2(config-if)# spanning-tree mst 2 port-priority 64
D2(config-if)# exit
D2(config)# interface range g1/0/5-6
```

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```
D2(config-if-range)# no shutdown
D2(config-if-range)# end
```

c. On A1, issue the **show spanning-tree mst 2** command and you will see that G1/0/4 is now the selected root port. This selection is based on the lower priority value of D2 interface G1/0/6. That lower priority value does not appear in any A1 output.

```
A1# show spanning-tree mst 2

##### MST2    vlans mapped:   4-5
Bridge        address 4464.3ce7.6880  priority      32770 (32768 sysid 2)
Root          address f8e9.4f9b.8700  priority      24578 (24576 sysid 2)
              port    Gi1/0/4         cost          20000     rem hops 19

Interface                     Role Sts Cost      Prio.Nbr Type
----------------              ---- --- --------- -------- --------------------
Gi1/0/1                       Altn BLK 20000     128.1    P2p
Gi1/0/2                       Altn BLK 20000     128.2    P2p
Gi1/0/3                       Altn BLK 20000     128.3    P2p
Gi1/0/4                       Root FWD 20000     128.4    P2p
```

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